Indian Institute of Technology Patna
Mid Semester Examination
Course Name: Large Language Models
Course Code: MA6119
Date: 29-9-2024

Maximum Marks: 40 Time: 2 Hours

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1. Consider a single-layer neural network with a sigmoid (σ) activation. If the input is a 3D vector $\mathbf{x} = [x_1, x_2, x_3]$, and weights are w_1, w_2, w_3, the output is given by $\sigma(w_1x_1 + w_2x_2 + w_3x_3 + b)$. What is the gradient of the output with respect to w_1? [Marks = 1]
   (a) $\sigma'(z) \cdot x_1$
   (b) $\sigma(z) \cdot (1 - \sigma(z)) \cdot x_1$
   (c) $\sigma(z) \cdot x_1$
   (d) $\sigma'(z)$

2. Which of the following best describes the update rule for the Adam optimizer?

$$\theta_{t+1} = \theta_t - \frac{\eta \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

In this equation, $\hat{m}_t$ and $\hat{v}_t$ represent: [Marks = 1]
   (a) The raw moment estimates for gradient and squared gradient
   (b) The bias-corrected first and second moment estimates
   (c) The exponentially decayed averages of gradient and squared gradient
   (d) The sum of all past gradients and squared gradients

3. In SGD with momentum, the velocity update is given by $v_{t+1} = \gamma v_t + \eta \nabla_\theta J(\theta_t)$, where γ is the momentum parameter. Which of the following describes the effect of the momentum term γv_t? [Marks = 1]
   (a) It accelerates gradient descent by accounting for past gradients
   (b) It reduces learning rate over time
   (c) It prevents overfitting by regularizing gradients
   (d) It increases gradient noise to encourage exploration

4. For which range of input values is the derivative of the sigmoid activation function $f(x) = \frac{1}{1+e^{-x}}$ highest?[Marks = 1]
   (a) $x \to \infty$
   (b) $x = 0$
   (c) $x \to -\infty$
   (d) $x = 1$

$f'(x) = \boxed{\sigma(1-\sigma)}$

if $x \to -\infty$ $\sigma = 0$ $f'(x) = 0$

if $x \to \infty$ $f'(x) = 0$

$x \to 0$ $f'(x) = \infty$

5. The Tanh activation function $f(x) = \tanh(x)$ outputs values in the range: [Marks = 1]
   (a) $[0, 1]$
   (b) $(-\infty, \infty)$
   (c) $[-1, 1]$
   (d) $[-2, 2]$

6. In a softmax function, the probability p_i for class i is given by:

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

If z_i is large compared to other z_j's, which of the following statements is true? [Marks = 1]

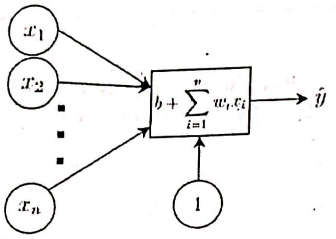
(a) p_i approaches 0

(b) p_i approaches 1

(c) p_i is undefined

(d) p_i does not change

7. Following is a tiny neural network. What kind of decision boundary can this model generate? [Marks = 1]



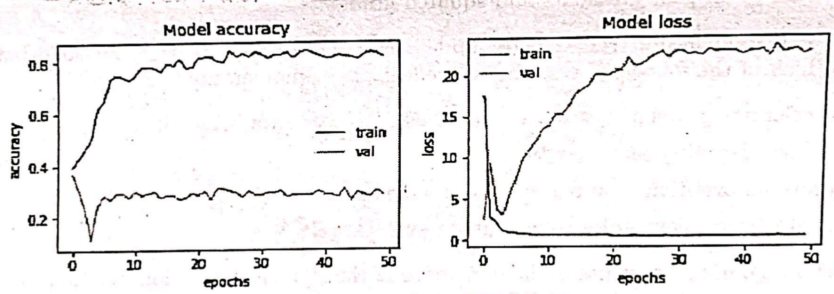
(a) The decision boundary is linear as there is only a single neuron in a single layer in this network.

(b) The decision boundary is linear as the output neuron does not apply a non-linear function on the linear transformation of the input.

(c) Irrespective of the size a neural network always generates a non-linear decision boundary.

(d) There is too little information to tell anything about the nature of the decision boundary of this network.

8. After examining the intriguing graphs depicting accuracy and loss over epochs during training and validation, what fascinating phenomenon is your model encountering? [Marks = 1]



(a) Underfitting

(b) Overfitting

(c) Convergence

(d) Class Imbalance

9. Compute the cross-entropy loss for a probability prediction vector [0.8,0.1,0.1] on a target vector of [1,0,0]. [Marks = 1]

10. Which one is the derivative of the ReLU (Rectified Linear Unit) activation function? [Marks = 1]

(a) ReLU is a non-differentiable function.

(b) 0 for all values of x.

(c) 1 for x > 0, 0 for x < 0.

(d) 1 for all values of x.

11. Suppose a certain fully connected neural network (FCNN) has a fully connected (FC) layer followed by a dropout layer. The output dimension of the FC layer is 200 and the dropout layer has the probability set to 0.4. What is the number of active neurons of the FC layer during training? [Marks = 1]

12. Consider a fully connected neural network (FCNN) with the following architecture designed for text classification:
Input: A 300-dimensional word embedding vector
Hidden layer 1: 512 output neurons, using ReLU activation
Hidden layer 2: 256 output neurons, using ReLU activation
Output layer: 5 neurons (for sentiment classification with 5 classes), using softmax activation
What is the total number of trainable parameters in this FCNN? [Marks = 2]

13. Explain the concept of dropout, describe how it is implemented during training versus testing, and discuss its impact on the performance of the model. **[Marks = 3]**

14. Explain how the K-Nearest Neighbors (KNN) algorithm works. How do the choice of the parameter K (the number of neighbors) and the selection of a distance metric (e.g., Euclidean or Manhattan distance) impact the model's performance? Additionally, outline the primary advantages and disadvantages of using the KNN algorithm. [Marks = 5]

15. (a) Derive the gradient of the Tanh and Leaky ReLU activation functions. [Marks = 3]

(b) Discuss the impact of these gradients on the training of deep networks, particularly focusing on the vanishing gradient problem for Tanh and the exploding gradient problem for Leaky ReLU. [Marks = 2]

16. Given the following loss function $L(\Theta)$, describe how the parameter update rule differs between Stochastic Gradient Descent (SGD), SGD with Momentum, RMSProp and the Adam optimizer. Write the mathematical update equations for each optimizer and explain how their differences affect convergence during training. [Marks = 5]

17. Consider a simple fully connected neural network with one hidden layer. The input is a 3-dimensional vector $\mathbf{x} = [x_1, x_2, x_3]$, and the network has 2 hidden units with ReLU activations. The output is a single scalar y. The network parameters include weight matrices $W_1 \in \mathbb{R}^{2\times3}$ and $W_2 \in \mathbb{R}^{1\times2}$, and bias vectors $\mathbf{b}_1 \in \mathbb{R}^2$ and $b_2 \in \mathbb{R}^1$.
The output of the network is given by:

$$z_1 = W_1\mathbf{x} + \mathbf{b}_1$$
$$a_1 = \text{ReLU}(z_1)$$
$$y = W_2 a_1 + b_2$$

Given the loss function $L = \frac{1}{2}(y - \hat{y})^4$, where $\hat{y}$ is the true label, derive the gradient of the loss function with respect to the parameters W_1, W_2, $\mathbf{b}_1$, and b_2. [Marks = 6]

18. In what ways does the choice of weight initialization influence the training process of a neural network? Provide an example to illustrate its effects on convergence speed and model performance. [Marks = 3]

$200 \times \frac{4}{10} = 80$

$1 + (x+h) + \frac{(x+h)^2}{2!} + \dots - \{1 - (x+h) + \frac{(x+h)^2}{2!} \dots\}$

$= \frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}$

$\frac{h}{\tan(x+h) - \tan x}$

Indian Institute of Technology Patna
End Semester Examination
Course Name: Large Language Models
Course Code: MA6119
Date: 30-11-2024

Maximum Marks: 60 **Time: 3 Hours**

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1. **Fully Connected Neural Networks** [Total Marks = 13]

(a) Given a text classification task with 10,000 unique words in the vocabulary, you design a neural network with the following fully connected layers:

- **Input layer**: 300-dimensional word embeddings.
- **Hidden layer 1**: 128 units, ReLU activation.
- **Hidden layer 2**: 64 units, ReLU activation.
- **Output layer**: 5 units (for 5 classes), softmax activation.

If the model uses a fully connected architecture, calculate the total number of trainable parameters (weights and biases) in the network. [Marks = 2]

(b) How does splitting a dataset into train, val and test sets help identify underfitting? [Marks = 2]

(c) You are designing a deep learning system to detect hate speech in social media posts. It is crucial that your model identifies as much hate speech as possible to prevent harmful content from spreading. Which of the following is the most appropriate evaluation metric: Accuracy, Precision, Recall, Loss Value? Explain your choice. [Marks = 2]

(d) You have a single hidden-layer neural network for a binary text classification task, where the input is $X \in \mathbb{R}^{n \times m}$ representing a bag-of-words (BoW) or word embeddings for m input samples, and the output $\hat{y} \in \mathbb{R}^{1 \times m}$ represents the predicted class probabilities for each sample. The true label is $y \in \mathbb{R}^{1 \times m}$. The forward propagation equations are as follows:

$$z^{[1]} = W^{[1]}X + b^{[1]}$$
$$a^{[1]} = \sigma(z^{[1]})$$
$$\hat{y} = a^{[1]}$$

The cost function is:

$$J = -\sum_{i=1}^{m} \left[y^{(i)} \log(\hat{y}^{[i]}) + (1 - y^{(i)}) \log(1 - \hat{y}^{[i]}) \right]$$

Write the expression for $\frac{\partial J}{\partial W^{[1]}}$ as a matrix product of two terms. [Marks = 4]

(e) Consider a neural network encoder $z = \text{softmax}[f_\theta(X)]$. You can think of f_θ as an MLP for this example. z is the softmax output, and we want to discretize this output into a one-hot representation before passing it into the next layer.

Consider the operation one_hot, where one_hot(z) returns a one-hot vector with a 1 at the argmax location. For example:

$$\text{one\_hot}([0.1, 0.5, 0.4]) = [0, 1, 0].$$

Say we want to pass this output to another fully connected layer g_ϕ to get a final output y.

(i) Is there a problem with the neural network defined below? [Marks = 1]

$$y = g_\phi(\text{one\_hot}(\text{softmax}(f_\theta(X))))$$

(ii) Consider the following function:

$$z = S_\tau(f_\theta(X)) = \text{softmax}\left(\frac{f_\theta(X)}{\tau}\right)$$

Here dividing by τ means every element in the vector is divided by τ. Obviously, when $\tau = 1$, this is exactly the same as the regular softmax function. What happens when $\tau \to \infty$? What happens when $\tau \to 0$?
Hint: You don't need to prove these limits; just showing a trend and justifying is sufficient. [Marks = 2]

[Total Marks = 21]

2. **RNNs, LSTMs and Transformer**

(a) Consider a 3-layer RNN where:

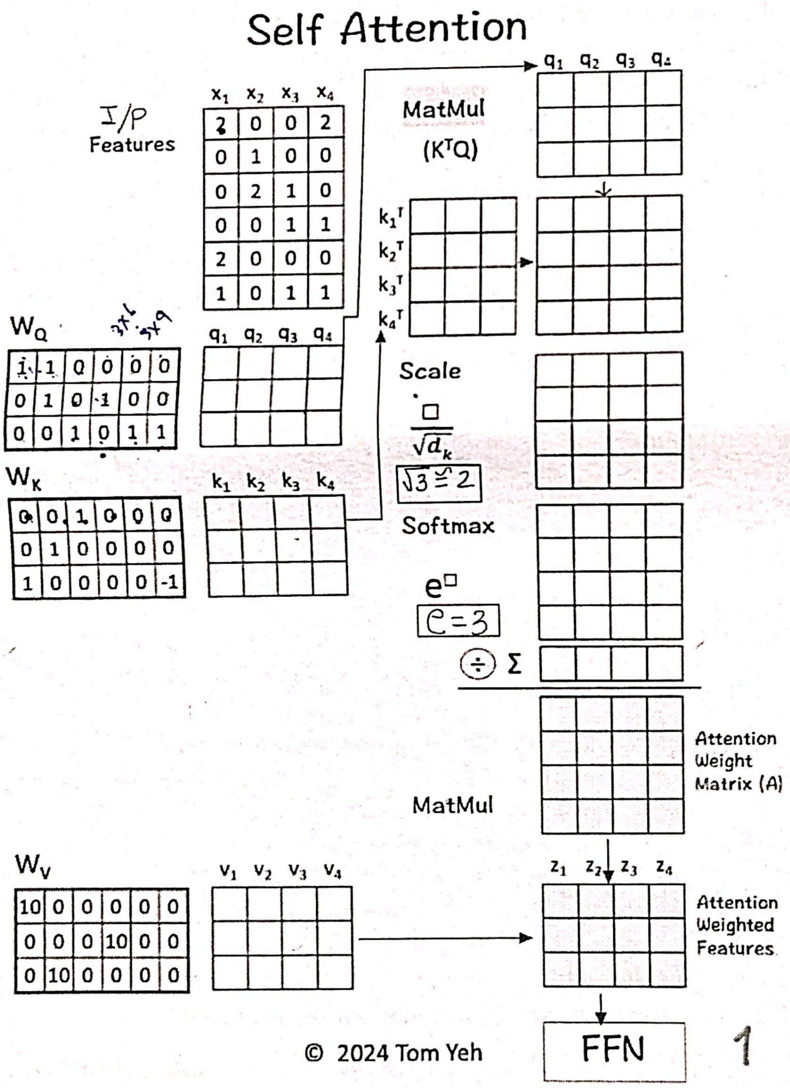
- The input dimension is 6.
- Each layer has 8 hidden units.
- The output dimension is 5.

How many trainable parameters (weights and biases) are there in total across all three layers? [Marks = 3]

(b) What challenges do Recurrent Neural Networks (RNNs) face during training due to the behavior of gradients, and how do these issues affect the learning process over long sequences? [Marks = 2]

(c) Provide the equations for a standard Long Short-Term Memory (LSTM) cell and explain the role of each gate (input gate, forget gate, output gate) and the cell state. [Marks = 3]

(d) Perform the right operations and fill in the blanks: [Marks = 6]



(e) You are tasked with building a model to predict daily stock prices based on historical data. The data exhibits strong temporal dependencies, with short-term fluctuations and occasional long-term trends. The dataset is relatively small, and each sequence has about 30 time steps. Which architecture (RNN, LSTM, or Transformer) would be most suitable for this task? Justify your choice considering the size of the dataset and the temporal dependencies. [Marks = 3]

(f) Design a Visual Question Answering (VQA) system where the input consists of an image and a natural language question, and the output is a text-based answer. Describe how you would extract features from the image and process the text input. How would you fuse the image and text features to generate the final answer? [Marks = 4]

3. **Word2Vec, BERT and GPT** [Total Marks = 15]

(a) How does Word2Vec differ from traditional one-hot encoding for word representation? [Marks = 2]

(b) What is the primary training objective of the Word2Vec model? How does it generate word embeddings? [Marks = 2]

(c) Explain the two main pre-training tasks of BERT: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). How do these tasks help in learning contextual representations? [Marks: 3]

(d) You are tasked with developing a sentiment analysis system for a product review platform to classify reviews as positive, negative, or neutral. Given the need to capture contextual meanings and handle nuanced language, you decide to use BERT. Explain how you would preprocess the review data for BERT, fine-tune the model for sentiment classification, and use it to predict sentiments for new reviews. Additionally, discuss the advantages of BERT in this context. [Marks: 3]

(e) What is GPT, and how does it generate human-like text based on the input it receives? [Marks: 3]

(f) Provide one use case where GPT outperforms BERT and explain why? [Marks: 2]

4. **ChatGPT and RAG** [Total Marks = 11]

(a) Explain how ChatGPT generates responses and describe the role of Reinforcement Learning with Human Feedback (RLHF) in fine-tuning the model. Compare RLHF with traditional supervised learning, using an example to highlight the differences in their training approaches. [Marks: 5]

(b) Explain the key components of the RAG pipeline and provide an example use case where it would be particularly effective. [Marks: 4]

(c) In RAG, if the retriever component fails to find relevant documents, what impact will this have on the generator's output? [Marks = 2]